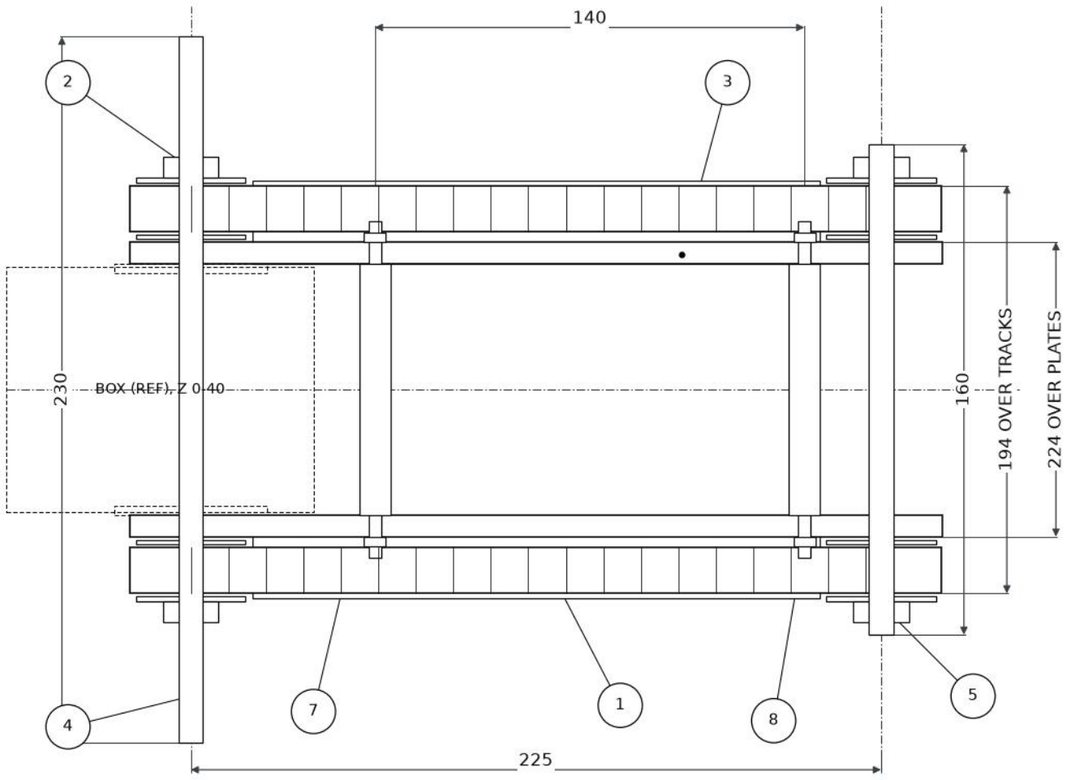
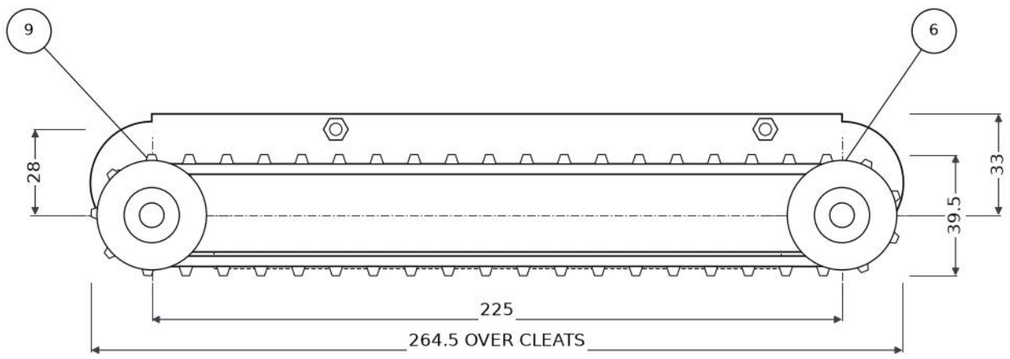
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	TS-01 TRACK	TRACK, TPU 95A, 110T HTD 5M, 45 CLEATS	2
2	TS-06 PULLEY_20T	PULLEY HTD 5M 20T, 15 MM BELT, BORE 8, BOUGHT	4
3	TS-02 SIDE_PLATE_FRONT	SIDE PLATE, FRONT SEGMENT, PETG, 1 L + 1 R	2
4	TS-04 AXLE_MIDDLE	MIDDLE AXLE, ROD 8 x 230, SHARED W/ REAR SEG	1
5	TS-04 AXLE_FRONT	FRONT AXLE, ROD 8 x 160	1
6	TS-07 BEARING_608ZZ	BALL BEARING 608ZZ, 8 x 22 x 7, BOUGHT	4
7	TS-05 SPACER_82	SPACER OD 10 x 82, BORE 4.5, PETG	2
8	M4_ROD_110	M4 THREADED ROD x 110 + 2 NUTS + 2 WASHERS	2
9	M3x14	M3 x 14 BOLT + NUT, HINGE SECTOR GEAR, BOUGHT	6

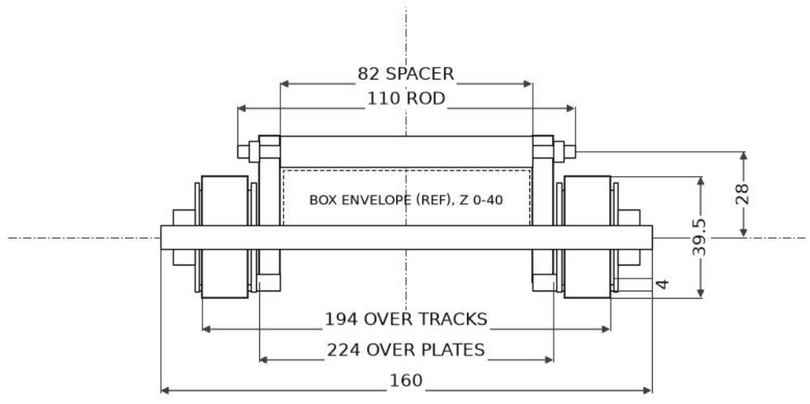
TOP VIEW



SIDE VIEW (FROM OUTBOARD)



END VIEW (FROM THE FRONT)



NOTES:
1. REAR SEGMENT: SAME, WITH ITS PLATES OUTBOARD (Z 105-112, TS-03), RAILS INBOARD, AND THE REAR AXLE (TS-04) INSTEAD OF THE FRONT AXLE.
2. BOTH SEGMENTS SHARE THE MIDDLE AXLE (ITEM 4), TURNED BY THE DRIVE MOTOR. PULLEYS ARE SET-SCREWED TO THE AXLES; THE AXLES TURN IN ITEM 6.
3. HINGE SECTOR GEAR (3 THICK, Z 38-41, NOT SHOWN) BOLTS TO THE INBOARD FACE OF EACH FRONT PLATE WITH ITEM 9. THE BOX STOPS AT Z 37 THERE.
4. THE BOX (TEAM'S) MUST END BEFORE X 50 SO THE FRONT CROSS BAR AT X 60 CLEARS IT WHEN THE HINGE BENDS. MM AT 1:5; X 5 FOR FULL SIZE.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
FRONT TRACK SEGMENT ASSEMBLY

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: SEE PARTS LIST
FINISH:

DWG NO.
TS-00

QTY.
1 (REAR SIMILAR)

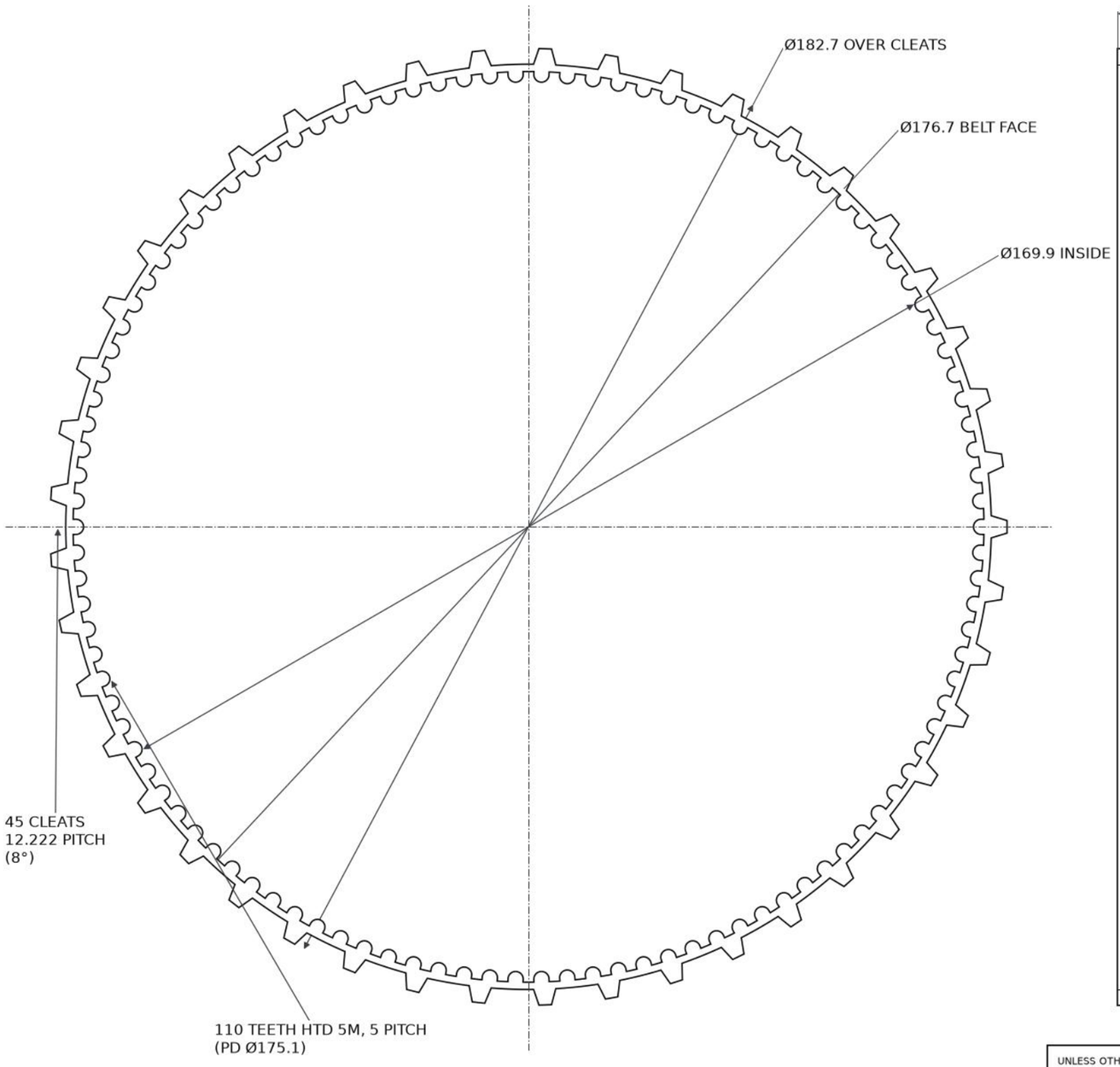
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE:
1:2

SHEET
1 OF 1

SIZE
A3

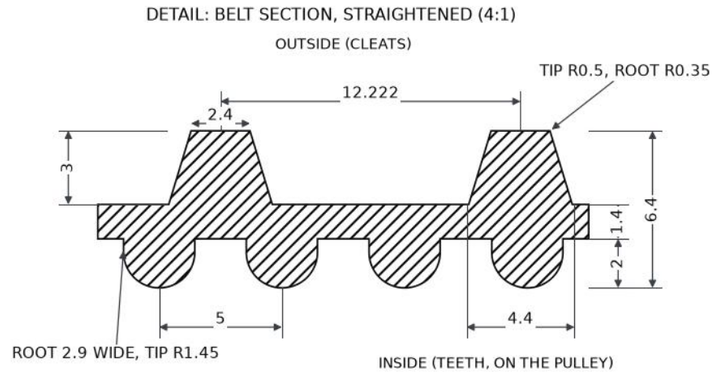
FRONT VIEW



EDGE VIEW



DETAIL: BELT SECTION, STRAIGHTENED (4:1)



NOTES:
1. FILE stl/track_TPU_x4.stl. PRINT STANDING AS A RING, NO SUPPORTS, 0.2 LAYERS, 100% INFILL, 20-30 MM/S, DIRECT-DRIVE EXTRUDER.
2. LOOP 550 ON THE PITCH LINE = 110 TEETH x 5 PITCH; RUNS ON TWO HTD 5M 20T PULLEYS (TS-06) AT 225 CENTRES. ABOUT 32 G.
3. 45 CLEATS AT 12.222 PITCH: 5 CLEATS = 61.11, ONE ES1 STEP ALONG THE SLOPE (35 x 50 AT 1:5), SO EVERY STEP EDGE SITS BETWEEN TWO CLEATS.
4. BED SMALLER THAN 190 MM: ASK FOR THE TWO-PIECE TRACK. PRINT THE 5 MM FIT-TEST SLICE FIRST (stl/fit_test_track_5mm_TPU.stl).

THIRD ANGLE



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
TRACK, 110T HTD 5M, 45 CLEATS

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: TPU 95A, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-01

QTY.: 4

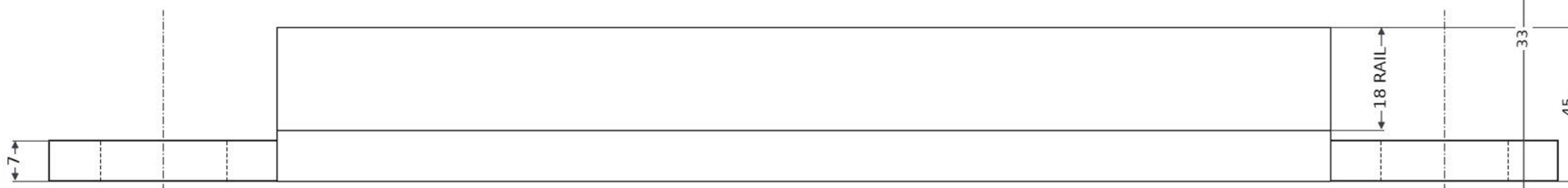
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 1:1, DETAIL 4:1

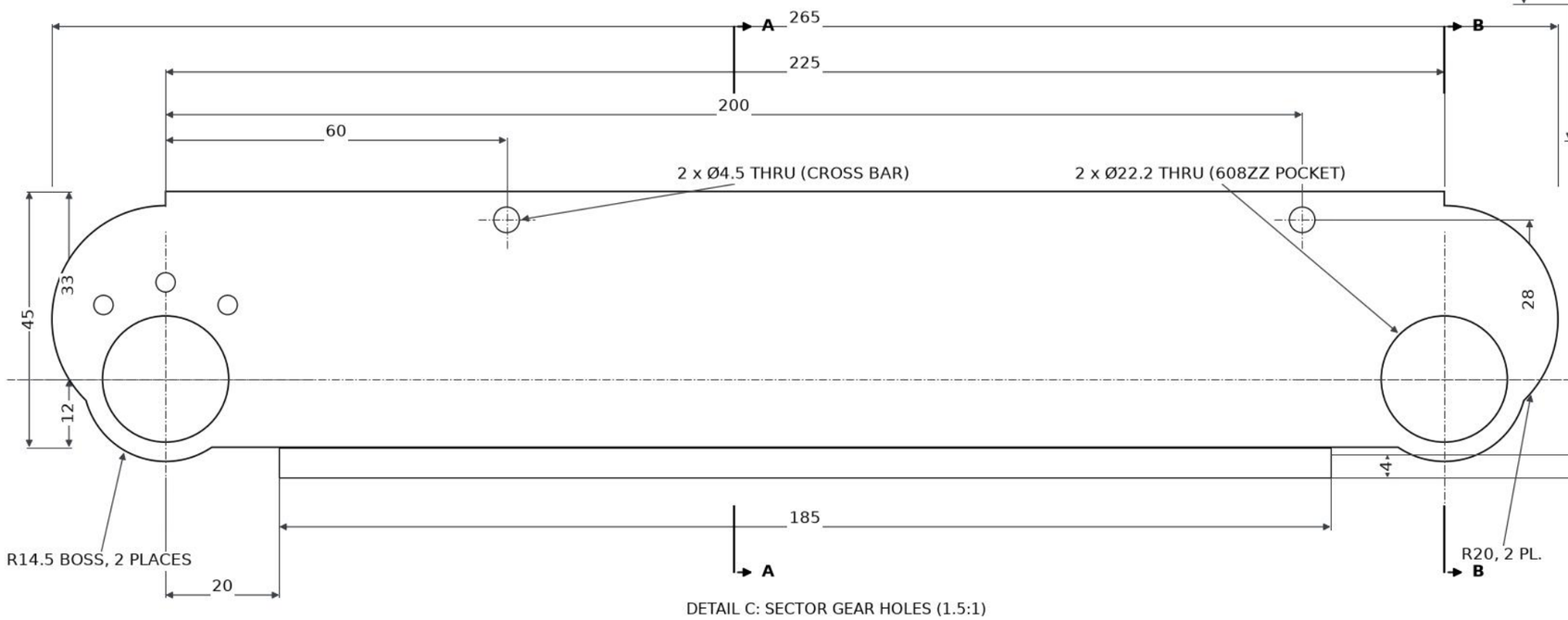
SHEET: 1 OF 1

SIZE: **A3**

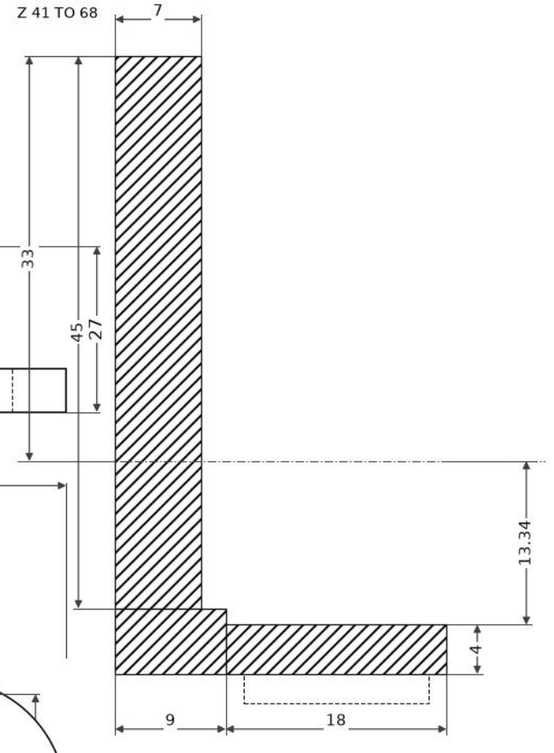
TOP VIEW



FRONT VIEW (FROM OUTBOARD)

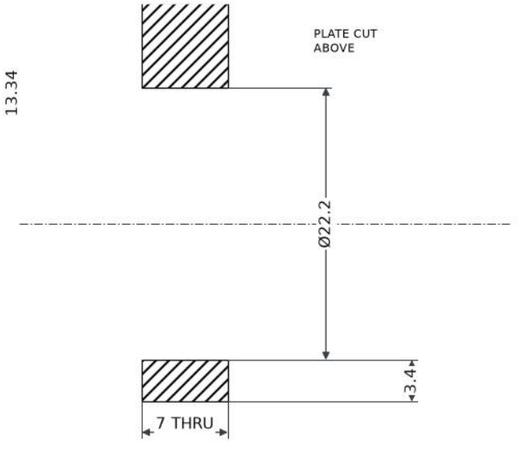


SECTION A-A (2:1)

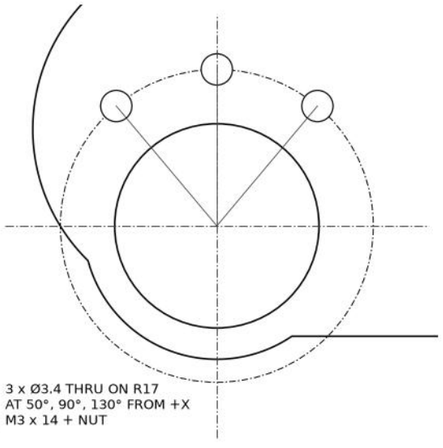


TRACK LOWER RUN (REF)

SECTION B-B (2:1)



DETAIL C: SECTOR GEAR HOLES (1.5:1)



3 x Ø3.4 THRU ON R17
AT 50°, 90°, 130° FROM +X
M3 x 14 + NUT

THIRD ANGLE

NOTES:
1. LEFT PLATE SHOWN; THE RIGHT ONE IS ITS MIRROR. FILES stl/side_plate_front_L.stl, _R.stl.
2. PRINT LYING ON THE PLATE FACE, 0.2 LAYERS, 4 WALLS, 30% INFILL, NO SUPPORTS.
3. BED 235 MM (OR 225 DIAGONAL); ON A SMALLER BED ASK FOR THE TWO-PIECE PLATE.
4. Z FROM THE VEHICLE CENTRE LINE: PLATE 41-48, WEB 41-50, RAIL 50-68
5. (RAIL CENTRED UNDER THE TRACK LANE 49-69). R20 ENDS ARE CENTRED 10.5 ABOVE THE AXLE.
6. POCKETS Ø22.2 THRU TAKE A 608ZZ (TS-07), LIGHT PRESS: MEASURE THE FIRST PRINT,
7. REAM IF TIGHT, GLUE IF LOOSE. THE RAIL TOP CARRIES THE LOWER RUN'S TOOTH TIPS.
8. X = 0 IS THE MIDDLE AXLE (HINGE); 3 x M3 SECTOR GEAR HOLES ON R17 ROUND IT, INBOARD FACE.

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:

SIDE PLATE, FRONT SEGMENT

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: PETG, 3D PRINTED
FINISH: AS PRINTED

DWG NO. TS-02

QTY. 2 (1 L + 1 R)

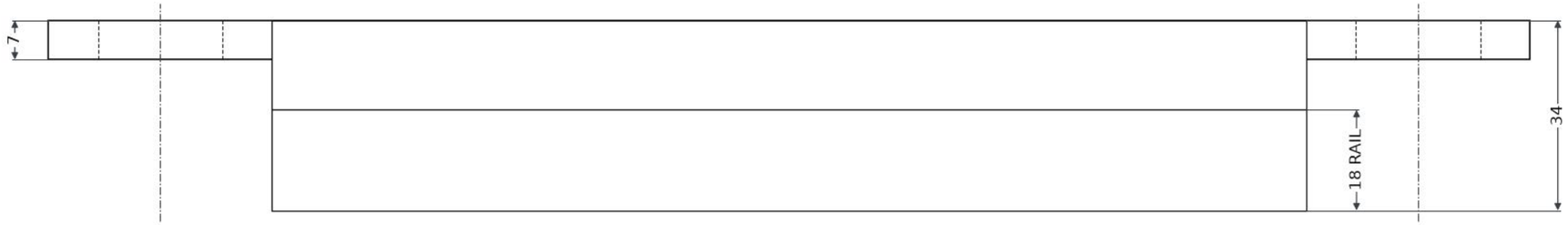
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 1:1, SECTIONS 2:1

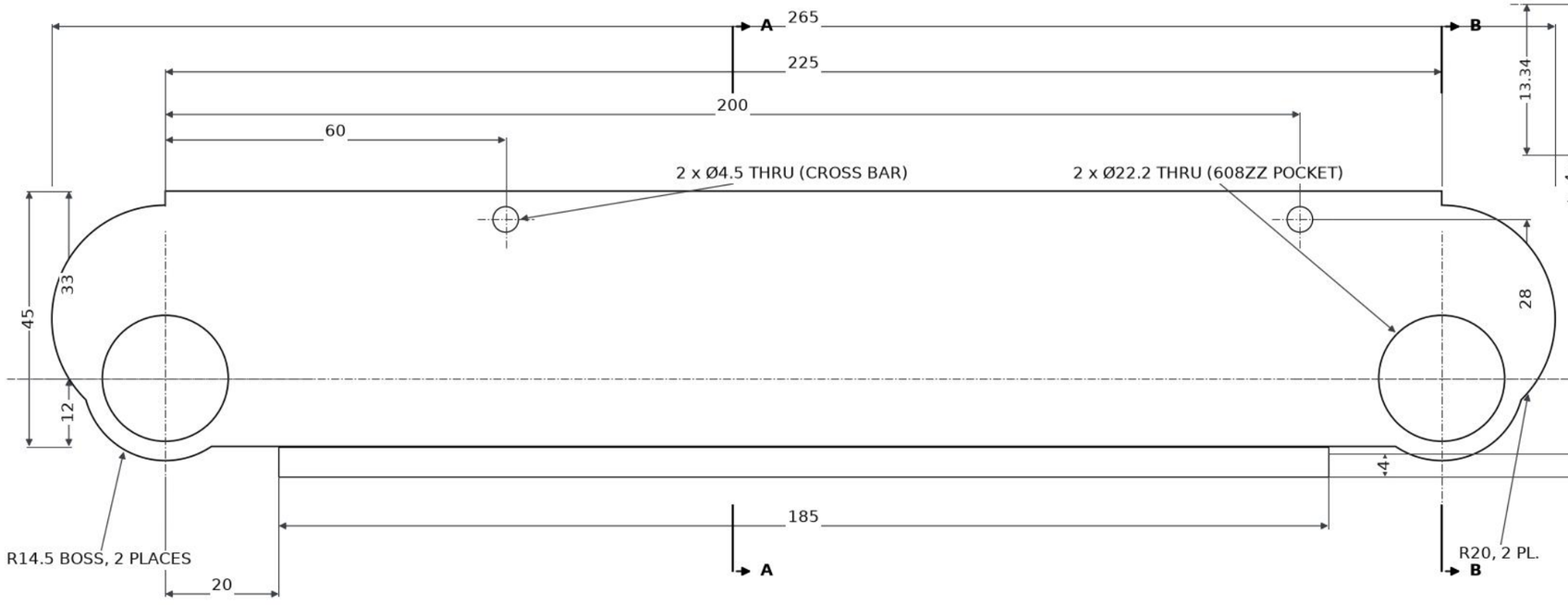
SHEET 1 OF 2

SIZE **A3**

TOP VIEW

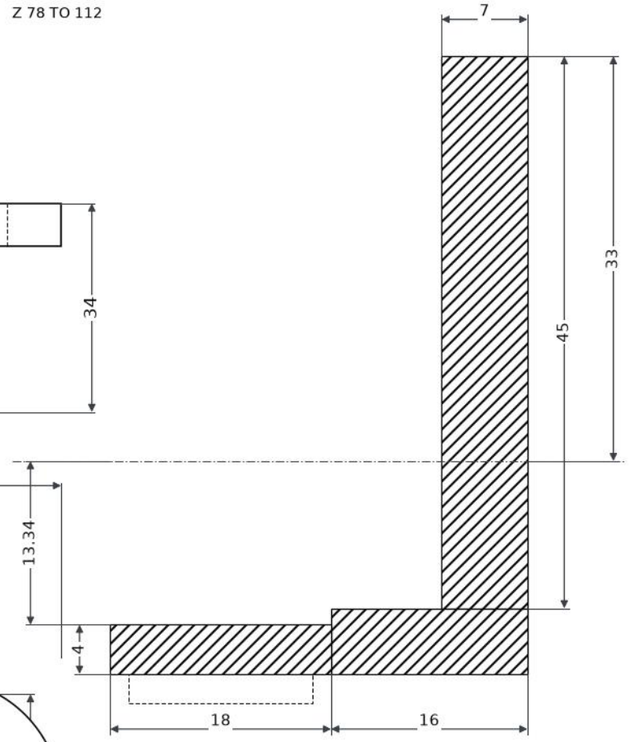


FRONT VIEW (FROM OUTBOARD)



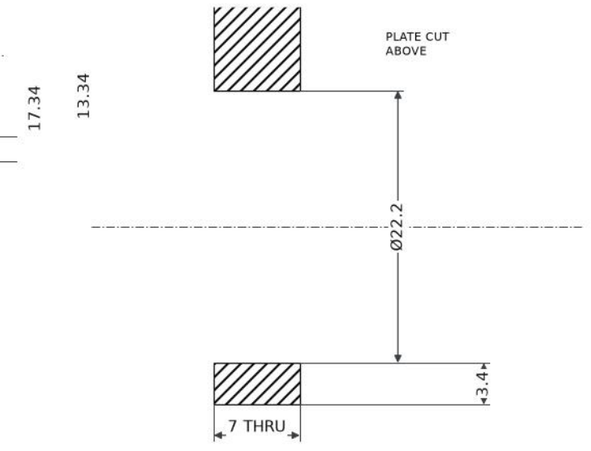
SECTION A-A (2:1)

Z 78 TO 112



TRACK LOWER RUN (REF)

SECTION B-B (2:1)



NOTES:
1. LEFT PLATE SHOWN; THE RIGHT ONE IS ITS MIRROR. FILES stl/side_plate_rear_L.stl, _R.stl.
2. PRINT LYING ON THE PLATE FACE, 0.2 LAYERS, 4 WALLS, 30% INFILL, NO SUPPORTS.
3. BED 235 MM (OR 225 DIAGONAL); ON A SMALLER BED ASK FOR THE TWO-PIECE PLATE.
4. Z FROM THE VEHICLE CENTRE LINE: PLATE 105-112, WEB 96-112, RAIL 78-96
5. (RAIL CENTRED UNDER THE TRACK LANE 77-97). R20 ENDS ARE CENTRED 10.5 ABOVE THE AXLE.
6. POCKETS Ø22.2 THRU TAKE A 608ZZ (TS-07), LIGHT PRESS: MEASURE THE FIRST PRINT,
7. REAM IF TIGHT, GLUE IF LOOSE. THE RAIL TOP CARRIES THE LOWER RUN'S TOOTH TIPS.
8. X = 0 IS THE REAR AXLE. THE BOX BOLTS TO THESE PLATES THROUGH THE Ø4.5 CROSS-BAR HOLES.

THIRD ANGLE



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
SIDE PLATE, REAR SEGMENT

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: PETG, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-03

QTY.: 2 (1 L + 1 R)

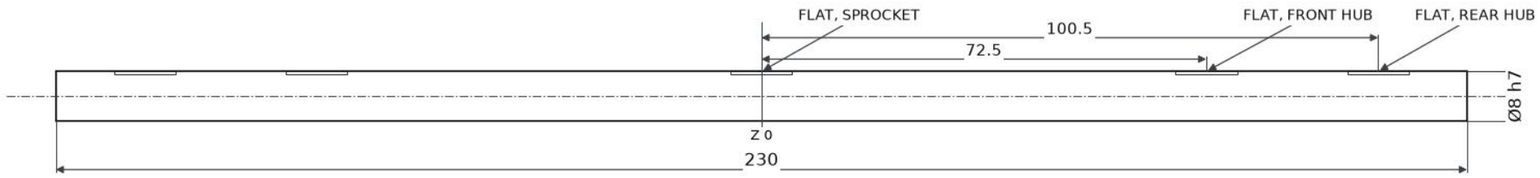
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 1:1, SECTIONS 2:1

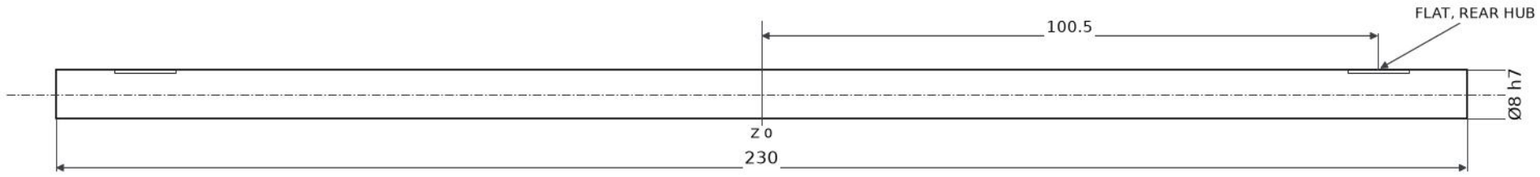
SHEET: 2 OF 2

SIZE: **A3**

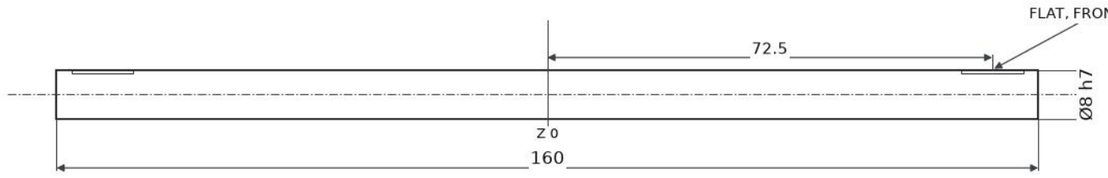
MIDDLE AXLE (HINGE, DRIVEN), QTY 1



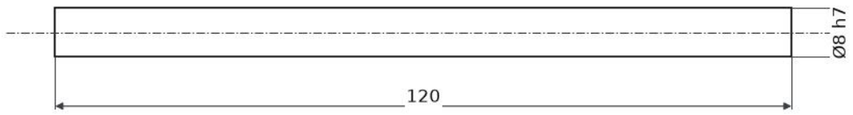
REAR AXLE, QTY 1



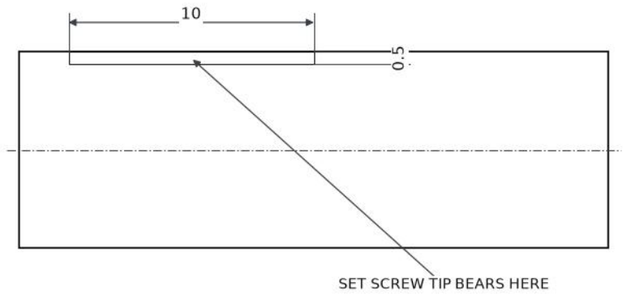
FRONT AXLE, QTY 1



HINGE CROSS SHAFT, QTY 1



DETAIL: FLAT UNDER A SET SCREW (4:1)



NOTES:
1. BUY: MISUMI 8 MM LINEAR SHAFT (UNHARDENED, SO THE FLATS CAN BE FILED) OR DESIGN LAB SPARE ROD. CUT TO LENGTH, ENDS SQUARE, DEBURRED: THE 608ZZ BORES ARE A CLOSE FIT.
2. FLATS 0.5 DEEP x 10 LONG UNDER EVERY PULLEY SET SCREW (AT THE HUB CENTRES, BOTH SIDES) AND UNDER THE DRIVE SPROCKET ON THE MIDDLE AXLE (Z 0).
3. THE MIDDLE AXLE IS THE HINGE: BOTH SEGMENTS' PLATES TURN ON IT IN THEIR 608ZZ. Z = 0 IS THE VEHICLE CENTRE LINE; THE AXLES ARE SYMMETRICAL ABOUT IT.
4. HINGE CROSS SHAFT: SAME ROD, 120 LONG, CARRIES THE TWO HINGE PINIONS ACROSS THE BOX IN 2 x 608ZZ; LENGTH TO BE CONFIRMED WITH THE HINGE GEAR DRAWING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
AXLES AND HINGE CROSS SHAFT

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: STEEL ROD Ø8 h7, UNHARDENED
FINISH: CUT SQUARE, DEBURR

DWG NO.: TS-04

QTY.: 3 + 1

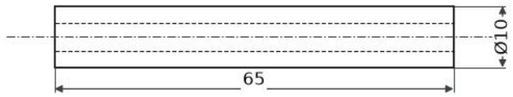
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 1:1, DETAIL 4:1

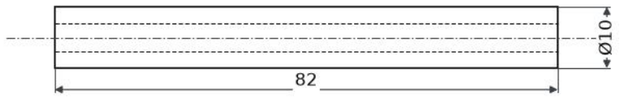
SHEET: 1 OF 1

SIZE: **A3**

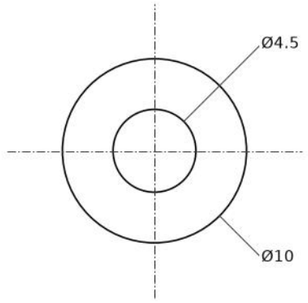
REAR SEGMENT SPACER, QTY 4 (2 PER ROD)



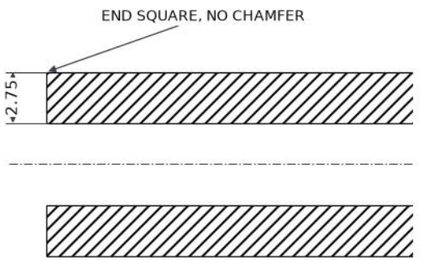
FRONT SEGMENT SPACER, QTY 2



END VIEW (3:1)



SECTION, ONE END (3:1)



NOTES:
1. FILES stl/spacer_front_x2.stl (82) AND stl/spacer_rear_x4.stl (65). PRINT STANDING ON END, 0.2 LAYERS, 3 WALLS, 20% INFILL.
2. FRONT: ONE 82 SPACER PER ROD BETWEEN THE FRONT PLATES (2 x 41). REAR: TWO 65 SPACERS PER ROD, ONE EACH SIDE BETWEEN THE BOX WALL (Z 40) AND THE REAR PLATE (Z 105).
3. THE NUTS OUTSIDE THE PLATES PULL PLATE - SPACER - BOX - SPACER - PLATE TOGETHER (SEE TS-08). AT 28 ABOVE THE AXLE THE SPACER CLEARS THE UPPER RUN'S CLEAT TIPS BY 3.3.
4. BORE 4.5 IS A CLEARANCE FOR THE M4 ROD. ENDS SQUARE: THEY SET THE PLATE SPACING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
CROSS BAR SPACER

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: PETG, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-05

QTY.: 2 x 82 + 4 x 65

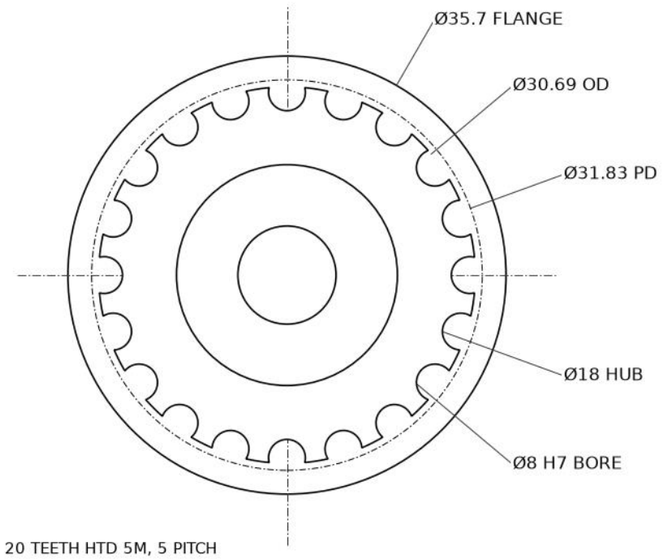
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 1:1, DETAIL 3:1

SHEET: 1 OF 1

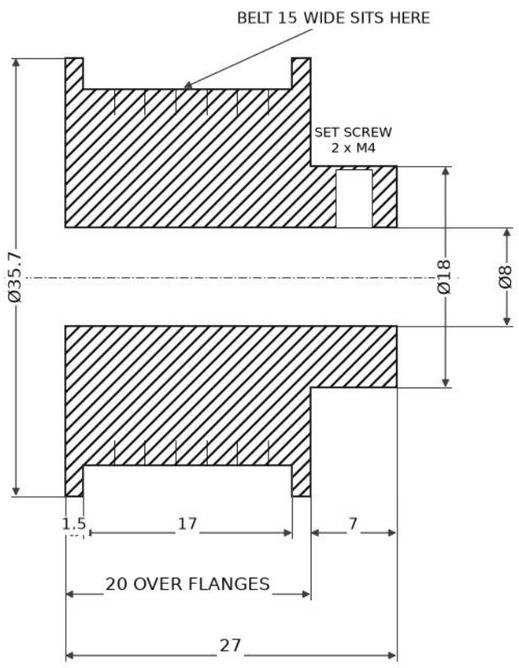
SIZE: **A3**

FRONT VIEW



20 TEETH HTD 5M, 5 PITCH

SECTION (ALONG THE AXLE)



NOTES:
1. BOUGHT PART. ORDER: HTD 5M, 20 TEETH, FOR 15 MM BELT, 8 MM BORE, FLANGES BOTH SIDES, SET-SCREW HUB. MISUMI SINGAPORE (APPROVED VENDOR). 4 PER SEGMENT.
2. WHAT MATTERS: 20T HTD 5M (PD 31.83), 15 MM BELT, 8 BORE. FLANGE, HUB AND LENGTH SHOWN ARE THE STAND-IN'S: CHECK THE VENDOR DRAWING.
3. LAYOUT ALLOWS 20 OVER THE FLANGES AND 7 FOR THE HUB (27 OVERALL) PER PULLEY. A LONGER PULLEY MOVES THE LANES: TELL THE TEAM BEFORE ORDERING.
4. PLA STAND-IN FOR FIT TESTS ONLY: stl/stand_in_pulley_20T_PLA_test_only.stl (M3 SET SCREW). NOT FOR DRIVING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
TIMING PULLEY HTD 5M 20T (BOUGHT)

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: ALUMINIUM (VENDOR'S)
FINISH: VENDOR'S

DWG NO.: TS-06

QTY.: 8

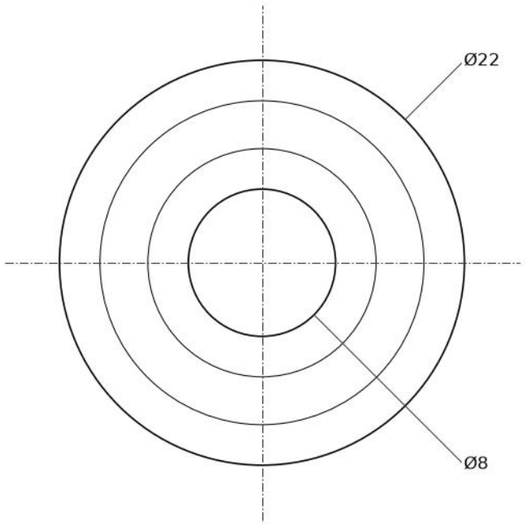
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 2:1

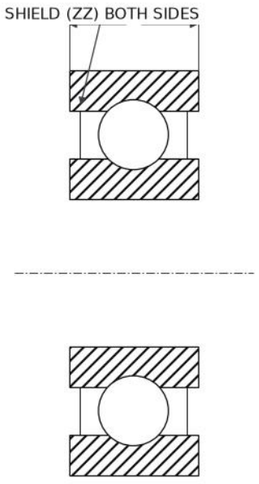
SHEET: 1 OF 1

SIZE: **A3**

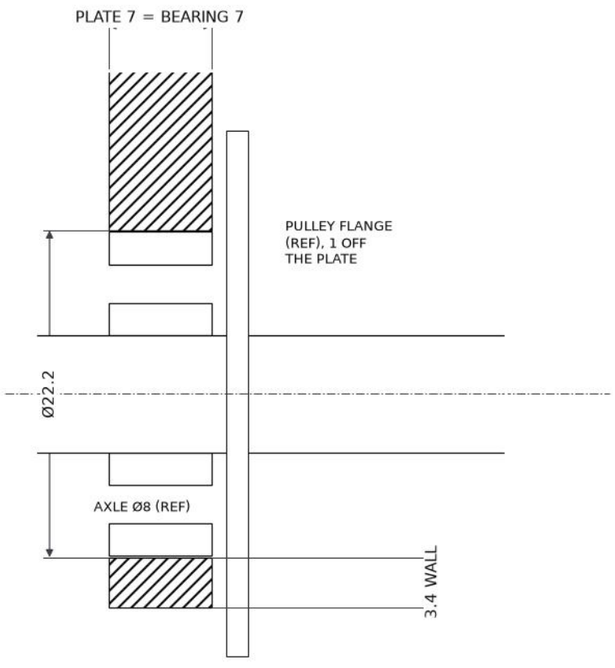
FRONT VIEW



SECTION



FITTED IN THE PLATE POCKET (2.4:1)



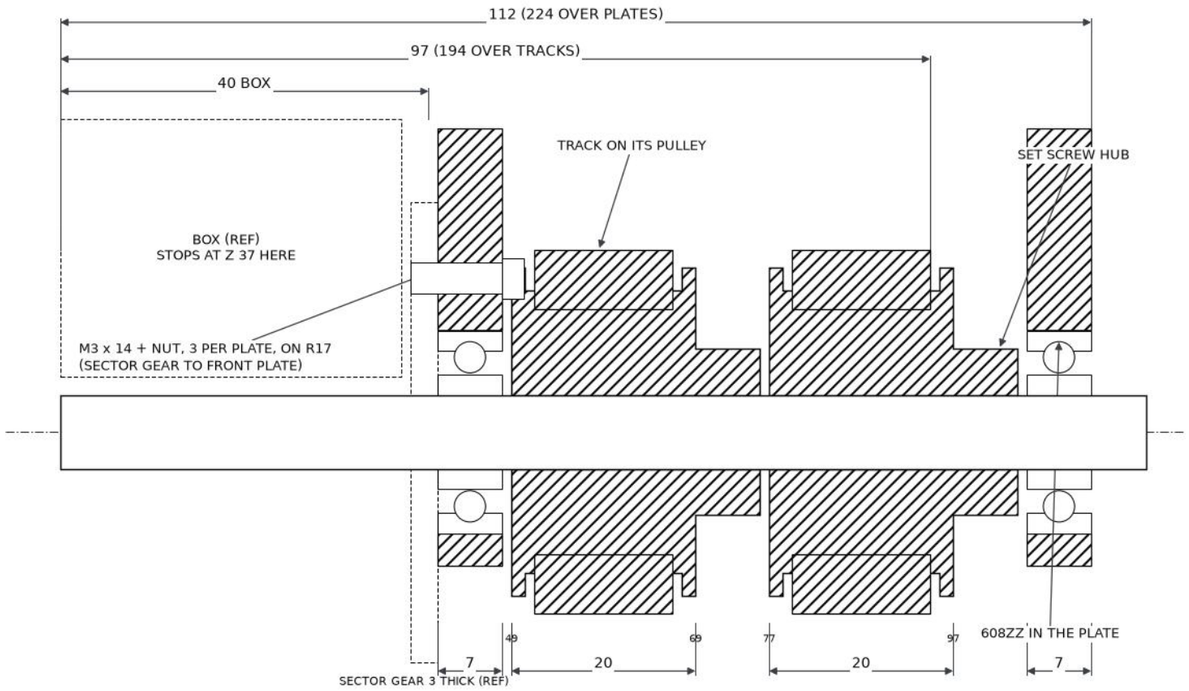
NOTES:
1. BOUGHT PART. 608ZZ, 8 x 22 x 7, SHIELDED BOTH SIDES. RS SINGAPORE (RS PRO 608-2Z) OR WHEELER MECTRADE / SKF (APPROVED VENDORS). ABOUT S$3 EACH.
2. QTY 10: 8 PRESSED INTO THE SIDE PLATE POCKETS (2 PER PLATE, TS-02 / TS-03), 2 IN THE BOX WALLS FOR THE HINGE CROSS SHAFT (TS-04).
3. POCKET Ø22.2 THRU IN THE PETG PLATE = OD + 0.2, A LIGHT PRESS. THE R14.5 BOSS GIVES A 3.4 WALL. MEASURE THE FIRST PRINT; REAM IF TIGHT, GLUE (CA) IF LOOSE.
4. REPLACES THE KFL08 FLANGE BLOCK ON THE QUOTATION SLIDE: 48 x 26 x 12 DOES NOT FIT IN THE 9 MM BETWEEN THE BOX (Z 40) AND THE FIRST TRACK (Z 49).

THIRD ANGLE

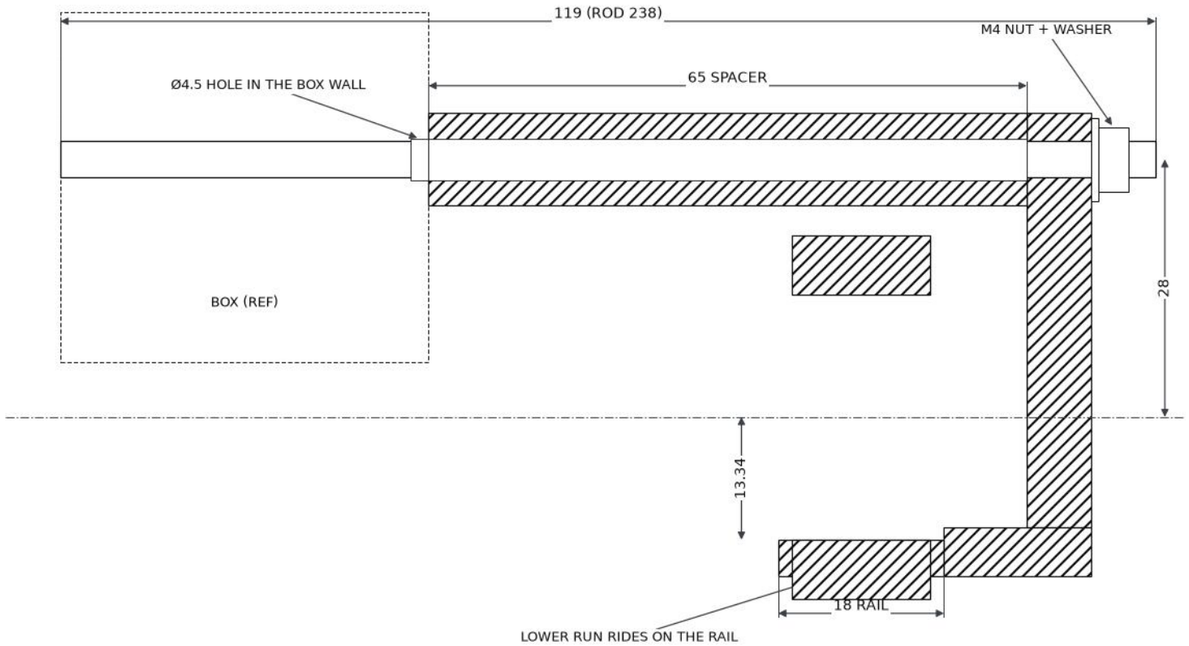


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS AT 1:5 MODEL SIZE TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG DO NOT SCALE DRAWING	TITLE: **BALL BEARING 608ZZ (BOUGHT)**		
DRAWN: NAZIM DATE: 2026-10-14			
MATERIAL: BEARING STEEL (VENDOR'S) FINISH: SHIELDED (ZZ)	DWG NO. TS-07	QTY. 10	
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE	SCALE: 3:1	SHEET 1 OF 1	SIZE **A3**

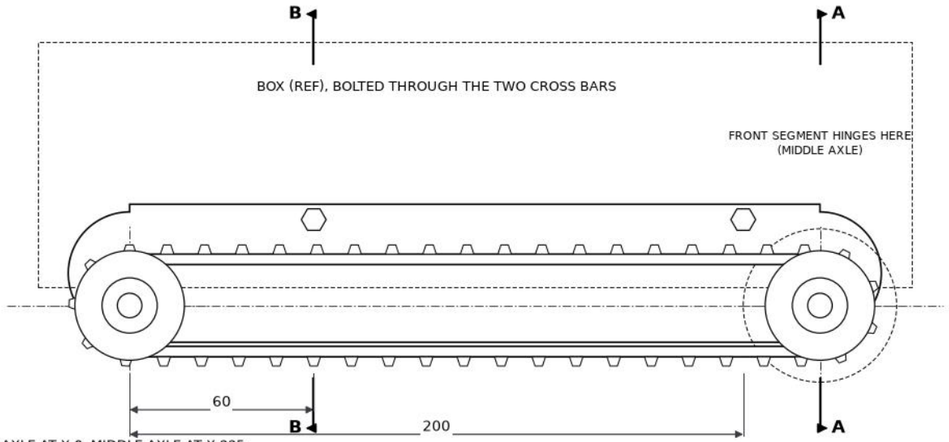
SECTION A-A THROUGH THE MIDDLE AXLE (HINGE), 1.5:1



SECTION B-B THROUGH A REAR CROSS BAR (X 60 FROM THE REAR AXLE), 1.5:1



SIDE VIEW, REAR SEGMENT ON THE BOX (FROM OUTBOARD), 1:2



NOTES:
1. ATTACHMENT: THE REAR SEGMENT BOLTS TO THE BOX BY ITS TWO CROSS-BAR RODS (M4 x 238), NUT + WASHER EACH END:
2. PLATE - 65 SPACER - BOX WALL - BOX WALL - 65 SPACER - PLATE. BOX NEEDS 2 x Ø4.5 HOLES PER WALL AT X 60 AND 200, 28 ABOVE THE AXLE.
3. THE FRONT SEGMENT IS HELD ONLY BY THE MIDDLE AXLE (HINGE) AND THE SECTOR GEARS, SO IT CAN SWING.
4. ONE SIDE SHOWN (Z FROM THE CENTRE LINE), THE OTHER MIRRORS. BOX, SECTOR GEAR AND PINION: TEAM'S PARTS, DASHED.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
REAR SEGMENT TO THE BOX

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: SEE TS-00
FINISH:

DWG NO. TS-08

QTY. 1

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 1.5:1 SECTIONS, 1:2 SIDE

SHEET 1 OF 1

SIZE **A3**

Parts list and build guide: track segments of the 1:5 prototype

Numbers from `parts_spec.json` v2 and the TS-00 to TS-07 sheets. Prices S$ ex GST as seen in vendor listings on 14 Oct 2026 (vendor pages could not be opened from this machine: confirm at checkout).

1. Does the track fit the school printers? Yes, in one piece.

Machine (NUS CDE)	Bed	Track ring Ø183 x 15, printed standing	Side plate 265 x 50 x 11, flat
Tiertime UP300, Fab Lab E4-02-06 (has a TPU extruder head)	205 x 255 x 225 mm	**fits** (183 wide, 183 tall)	**fits on the diagonal** (327 mm available; 265 needed)
Tiertime UP600, Fab Lab E4-02-06	400 x 500 x 600 mm	fits	fits straight
any hobby printer with a 190 mm bed or less		does not fit: ask for the two-piece track with links	ask for the two-piece plate

So: **no split, no links**, as long as the tracks go on the UP300 or UP600. Keep the two-piece version as a fallback only.

2. What is made where, and the one-line justification for the request form

The Professional Workshop (E3-01-06) is CNC mills, a turn-mill and wire EDM: metal parts, done by the staff from a drawing. The Fab Lab (E4-02-06) is the 3D printers and the laser cutter. Job requests go through the EDIC request form.

Part	Qty	Make how	Where	Justification (copy this)
Track	4	3D print, TPU 95A, standing, 100% infill, 0.2 mm, no supports. File `stl/track_TPU_x4.stl`	Fab Lab UP300 (TPU head)	A toothed flexible belt with step-matched cleats on the outside; no stock belt has the cleats, and the cleat pitch (12.22 mm) is set by the ES1 step size.
Side plate, front L and R, rear L and R	4	**Option A:** 3D print PETG, flat, 4 walls, 30% infill. Files `stl/side_plate_{front,rear}_{L,R}.stl`. **Option B:** CNC-mill from 7 mm aluminium plate to sheets TS-02 / TS-03 (the drawings are made for this).	A: Fab Lab UP600 or UP300. B: Professional Workshop, by request form	One part carries the two bearing pockets, the slide rail under the lower track run and all holes; a cut sheet would need the rail and bearing housings as extra parts.
Cross bar spacer, 82 (front) and 65 (rear)	2 + 4	3D print PETG (`stl/spacer_front_x2.stl`, `stl/spacer_rear_x4.stl`), or cut from Ø10 aluminium tube with a 4.5 bore on the lathe	Fab Lab, or Training Workshop with the plates	Non-standard lengths set the plate spacing; the rear ones sit between the box wall and the plate.
Stand-in pulley, PLA, test only	1	3D print PLA (`stl/stand_in_pulley_20T_PLA_test_only.stl`)	Fab Lab, Tinkering Corner or NUS Library	Fit test of the printed teeth before the bought pulleys arrive.

If printing is the hassle: **Option B for the plates** moves the biggest print (4 x 5 h) to the Professional Workshop staff, who machine it from the TS-02/TS-03 drawings. The tracks cannot be avoided: they must be printed in TPU. Everything else is bought.

3. Every piece, and which segment it belongs to

The prototype has two track segments: the **front segment** (hinged, plates inboard) and the **rear segment** (fixed to the box, plates outboard). They share the middle axle.

Front segment (sheet TS-00)

Item	Piece	Qty	Note
1	Track	2	one each side
2	Pulley HTD 5M 20T	4	2 each side: one on the middle axle, one on the front axle
3	Front side plate	2	1 left + 1 right (mirror images, not the same part)
6	Bearing 608ZZ	4	2 pressed into each plate
5	Front axle Ø8 x 160	1	
7	Spacer Ø10 x 82	2	one per cross bar
8	M4 threaded rod x 110	2	cut from the 1 m rod
8	M4 nut + M4 washer	4 + 4	one each end of each rod
9	M3 x 14 bolt + M3 nut	6 + 6	3 per plate, hold the hinge sector gears (team's part)
-	M4 set screw	8	2 per pulley; MISUMI does not include them

Rear segment

Item	Piece	Qty	Note
1	Track	2	
2	Pulley HTD 5M 20T	4	2 each side: middle axle and rear axle
-	Rear side plate	2	1 left + 1 right
6	Bearing 608ZZ	4	2 per plate
-	Rear axle Ø8 x 230	1	
-	Spacer Ø10 x 65	4	two per rod, one each side between the box wall (z 40) and the plate (z 105)
-	M4 threaded rod x 238	2	**these are what bolt the segment to the box**: plate - spacer - box wall - box wall - spacer - plate (sheet TS-08)
-	M4 nut + washer	4 + 4	one each end of each rod
-	M4 set screw	8	

Shared

Piece	Qty	Note
Middle axle Ø8 x 230 (item 4)	1	the hinge; carries 4 pulleys (2 front-segment, 2 rear-segment) and runs in all 4 plates'

		bearings
Hinge cross shaft Ø8 x 120	1	runs across the box, carries the hinge pinions (team's part)
Bearing 608ZZ	2	in the box walls, for the cross shaft

Totals to get

Piece	Total	Make / buy
Track	4	make
Pulley 20T	8	buy
Side plate	4 (2 front L/R + 2 rear L/R)	make
Bearing 608ZZ	10	buy (2 packs of 5)
Ø8 rod, cut to 230 + 230 + 160 + 120 = 740 mm	2 x 400 mm	buy (or Design Lab spare)
Spacer	6 (2 x 82, 4 x 65)	make
M4 threaded rod, cut to 110 + 110 + 238 + 238 = 696 mm	1 x 1 m	buy
M4 nut / M4 washer	8 / 8	buy or Design Lab spares
M3 x 14 bolt + nut	6 + 6	buy or spares
M4 set screw (M4 x 4 or x 5, cup point)	16	buy with the pulleys
Stand-in pulley, PLA	1	make

4. How it goes together (one segment; the other is the same)

1. **Bearings into plates.** Press two 608ZZ into each plate's Ø22.2 pockets (a vice with wood packing, or a 22 mm socket as a drift). Flush with the plate face.
2. **Pulleys onto the axles.** Slide each pulley onto its axle, hub facing outboard (away from the plate), and set-screw it on the filed flat at the z station on sheet TS-04. On the middle axle the order from the centre out, each side, is: front plate bearing (z 41-48), front-segment pulley (49-69, hub 69-76), rear-segment pulley (77-97, hub 97-104), rear plate bearing (105-112).
3. **Track on, then the second plate.** Fit the left plate onto the middle and front axles. Loop a track over the two left pulleys (TPU stretches a little; walk it on like a bicycle chain). Then slide the right plate on and loop the right track. Doing the track before the outer plate is what makes it possible.
4. **Cross bars.** Front segment: M4 rod through one plate, the 82 spacer, the other plate; washer and nut each end. Rear segment: the same rods also hold the box. Rod through the plate, a 65 spacer, through the box wall, across the box, out the other wall, the second 65 spacer, the other plate; washer and nut each end, tightened until plates, spacers and box are clamped as one. Two per segment, at x 60 and 200 from the rear axle, 28 above the axle line. The box needs two Ø4.5 holes through each wall at those positions (sheet TS-08).
5. **Check:** turn the middle axle by hand, both tracks run without clicking; the lower run rests on the slide rails.
6. **Hinge (team's parts).** Sector gears bolt to the inboard face of the front plates with the six M3 x 14 and nuts (7 plate + 3 gear + nut). The front segment is held to the box only by the middle axle and these gears, so it can swing; nothing else is bolted.

5. Buy list with links

Buy	Spec	Where	Price	Qty	Arrives

Pulley	HTD 5M, 20 T, 15 mm belt, Ø8 bore, flanged, set-screw hub (MISUMI economy family C-HTPA20H5M150, bore 8)	MISUMI SG, Timing Pulleys HTD 5M	~S$20 each (quote)	8	ships 1 day from SG stock per the catalogue; allow 2-5 working days
Set screws	M4 x 4 cup point (MISUMI MSSU4-3 or equal)	same order, or Design Lab	~S$0.20	16	with the pulleys
Bearing	608ZZ (608-2Z/C3), 8 x 22 x 7, shielded	RS SG pack of 5, 261-2603	S$14.82 per pack	2 packs	next working day (order before 3:30 pm), max 4 working days
Ø8 rod	8 mm x 400 mm steel shaft; hardened, so grind the flats (Dremel), or buy soft rod from a hardware shop on the vendor list	Design Lab spare first; else RS SG linear shafts	S$37 each at RS; ~S$5 at a hardware shop	2	next working day / same day
M4 rod	stainless M4 x 1 m, DIN 975	RS SG 200-9958 (pack of 5, S$49.69) or one rod from a hardware shop	S$10 per rod	1	4 working days / same day
Nuts, washers, bolts	M4 nuts x 8, M4 washers x 8, M3 x 14 + nuts x 6	Design Lab spares; else Element14 SG nuts and RS SG	S$0 to S$15	22	1-2 business days
TPU filament	TPU 95A, 1.75 mm, 1 kg (4 tracks use about 130 g)	Element14 SG, Multicomp Pro, or the Fab Lab's own stock	~S$45	1	1-2 business days

Totals: about S$160 of Singapore-priced parts (bearings, rod, M4 rod, nuts), about S$45 with Design Lab spares, plus the pulleys (~S$160, quote) and TPU (~S$45). Whole set roughly **S$300-370 ex GST**. RS: free delivery over S$150 ex GST on a private account, else S$25 handling.

Order of events: day 0 order everything and print the stand-in pulley + fit-test slice; day 1-2 fit test, print or request the plates; day 2-5 parts arrive, press bearings, cut rod; day 3-6 print tracks, assemble one segment, run it on the test stairs. The pulleys are the only item that can hold the build up, so order them first.

Motion and drive

, and how they link to the motor and the box

All of this is in `parts_spec.json` (`layout_decision`); it is repeated here because the drawings show the parts one at a time.

- **Every pulley is set-screwed to its axle** (a filed flat under each set screw). The **axles turn**

in the 608ZZ bearings pressed into the plates; the plates themselves do not turn with the axle.

- **One motor drives everything.** The drive motor turns the **middle axle** through a sprocket that

sits on it between the two front plates, inside the box (the box ends at z = 40 each side; the front plates are at z = 41 to 48). Because the middle axle carries the rear pulley of the front segment and the front pulley of the rear segment, both segments' tracks are driven from it. The rear and front axles are idlers: they turn because the track turns their pulleys.

- **The middle axle is also the hinge.** The front segment pivots about it on its plate bearings.

The hinge sector gear bolts to the rear end of each front plate (3 x M3 on a 14 mm radius); its pinion sits on the **hinge cross shaft**, which runs across the box in the two spare 608ZZ. Turning the cross shaft turns the sector gears, so the front segment bends up or down about the middle axle while the tracks keep running. The sector

gear, pinion and hinge motor are outside this drawing set.

- **The rear segment is fixed to the box.** Its outboard plates (z = 101 to 108) carry 2 x M4 holes

at x = -60 and -200 from the middle axle, 22 mm above the axle line, for the box cross bars (4 x M4 x 12). The front segment's plates are tied to each other only by their two cross bars (M4 rod + 82 mm spacer) so the segment can swing.

- **The lower run of each track rides on the plate's slide rail** (top face 13.34 mm below the axle),

so the track cannot sag between the pulleys when it is on the steps.